MIL-S-14374C
11 February 1983
SUPERSEDING
MIL-S-14374B(MU)
2 October 1969

MILITARY SPECIFICATION

SILICA GEL, IRON FREE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one iron-free type of silica gel (hereinafter referred to as silica gel).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

FSC 6810

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:07Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

C169 - Chemical Analysis of Soda-Lime and Borosilicate Glass
D1193 - Reagent Water
D1895 - Apparent Density, Bulk Factor, and Pourability of Plastic Materials
E11 - Wire-Cloth Sieves for Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Silica gel shall conform to the chemical and physical characteristics of table I when tested as specified therein.

3.2 Particle size. Silica gel shall have a particle size distribution conforming to table II when tested as specified in 4.2.4.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:07Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement	Test paragraph
Silicon dioxide, percent by weight, minimum	99.7	4.2.4.1
Apparent density, pounds per cubic foot	37 to 43	4.2.4.2
Iron, parts per million, maximum	20	4.2.4.3
Chloride, parts per million, maximum	2	4.2.4.4
Moisture, percent by weight, maximum	3.0	4.2.4.5
Contamination	To pass test	4.2.4.6
pH	6 to 7	4.2.4.7

TABLE II. Particle size distribution

Passing sieve (micrometers)	Retained on sieve (micrometers)	Percent by weight	
		Minimum	Maximum
---	600	--	0
600	425	--	2
425	250	80	--
250	180	--	20
180	150	--	3
150	---	---	3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the silica gel produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:07Z
Check the source to verify that this is the current version before use.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table III. A representative specimen of approximately 125 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Unitization not as specified
(f) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 type III and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Silicon dioxide. Determine the percent silicon dioxide in accordance with ASTM C169, using the procedure which determines percent silicon dioxide.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:07Z
Check the source to verify that this is the current version before use.

4.2.4.2 Apparent density. Determine the apparent density in accordance with ASTM D1895 and convert to pounds per cubic foot.

4.2.4.3 Iron. Weigh 2.000 g of the specimen into a 100-milliliter (ml) platinum evaporating dish. Add 2 ml of concentrated sulfuric acid and 10 ml of concentrated hydrofluoric acid, and evaporate slowly on a hot plate under a hood to fumes of sulfur trioxide. Continue the evaporation until the volume is approximately 0.7 ml. Remove from the hot plate, cool, slowly add an additional 10 ml of concentrated hydrofluoric acid, and repeat the evaporation to fumes of sulfur trioxide. Cool, add 10 ml of 4-percent boric acid solution, and evaporate to approximately 1 ml. Cool, quantitatively transfer the concentrate to a 50-ml volumetric flask, and make up to volume with water. Transfer a 2-ml aliquot of the solution to a color comparison tube and dilute to 50-ml with water. Add 30 to 50 milligrams (mg) of ammonium persulfate crystals and 3 ml of ammonium thiocyanate solution. The color developed shall be no darker than that produced in a control containing 0.01 mg of iron ions in an equal volume of solution containing the quantities of reagents used in the specimen test.

4.2.4.4 Chloride. Weigh 10 g of the specimen and dissolve in 50 ml of water. Bring the solution to a boil and continue boiling for 2 to 3 minutes. Decant 25 ml and to it add 1 ml of concentrated nitric acid and 1 ml of 0.1N silver nitrate solution. Any turbidity produced shall be no greater than that produced in a control containing 0.01 mg of chloride ion in an equal volume of solution containing the quantities of reagents used in the specimen test.

4.2.4.5 Moisture. Weigh to the nearest milligram approximately 2 g of the specimen into a tared weighing dish. Dry to a constant weight in an oven at 105° ± 5°C, cool to room temperature in a desiccator, and weigh to the nearest milligram. Calculate the percent by weight moisture as follows:

$$\text{Percent by weight moisture} = \frac{100\ (A - B)}{W}$$

where: A = Weight of dish and specimen before drying, in grams,
B = Weight of dish and specimen after drying, in grams, and
W = Weight of specimen in grams.

4.2.4.6 Contamination. Add 1 g of specimen to 50 ml of 10-percent sodium hydroxide solution. The specimen shall show no appreciable darkening after 5 minutes.

4.2.4.7 pH. Weigh 5.0 g of specimen into a glass-stoppered Erlenmeyer flask and dissolve in 50 ml of carbon dioxide- and ammonia-free water. Stopper the flask, shake occasionally, and let stand overnight. Determine the pH of the aqueous extract by means of the glass electrode method within 10 minutes after it is exposed to the air.

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:07Z
Check the source to verify that this is the current version before use.

4.2.4.8 Particle size. Use tared sieves conforming to ASTM E11. Arrange the sieves in the following ascending order: receiving pan, 150 micrometer, 180 micrometer, 250 micrometer, 425 micrometer, 600 micrometer, and cover. Place 100 ± 1 g of the specimen in the top sieve (600 micrometer), cover, and shake for 10 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the material retained on each sieve and the material on the receiving pan and calculate the percent by weight retained by each sieve and the percent by weight passing the 150 micrometer sieve as follows:

$$\text{Percent by weight retained on applicable sieve} = \frac{100\ (A - B)}{W}$$

$$\text{Percent by weight passing 150 micrometer sieve} = \frac{100\ (C - D)}{W}$$

where: A = Weight of applicable sieve and specimen retained, in grams,
B = Weight of applicable sieve, in grams,
C = Weight of receiving pan and specimen collected, in grams,
D = Weight of receiving pan, in grams, and
W = Weight of specimen in grams.

5. PACKAGING

5.1 Unit packing, level B. A quantity of one pound (+0.2 or -0 ounces) of silica gel shall be packed in a bottle conforming to group A, class 1, type optional, style 2, grade optional, closure A of PPP-C-186. The closure shall be provided with an aluminum foil liner. The filled and closed bottle shall be provided with an outer seal conforming to seal A of PPP-C-186. The bottles with closures in place and seals applied shall pass the leakage test specified in PPP-C-186.

5.2 Packing, industrial. Silica gel, unit packed as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Unitization. Uniform quantities of packs of silica gel shall be palletized in a manner to assure carrier acceptance, safe delivery to first destination, and stable stackability in storage.

5.4 Marking. Unit packs and packs of silica gel shall be marked in accordance with MIL-STD-1188. Pallet loads shall be marked to assure carrier acceptance and safe, timely delivery to the destination. Containers shall be marked to show date of manufacture and the lot number of the silica gel.

6. NOTES

6.1 Intended use. Silica gel is intended for use in the preparation of impregnated gels and in the manufacture of sampling tubes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:07Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodian:

Army - EA

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B376

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:07Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armement Research and Development Command
ATTN: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:07Z
Check the source to verify that this is the current version before use.